

Interpretable Multimodal Emotion Recognition on MELD with LLM-Generated Explanations

A Comparative Study of Fusion Architectures, Loss Functions, and Explanation Consistency

Muhammad Faizan Tanveer

MSc Computer Science and Engineering, University of Oulu
Student ID: 2509524 | Email: mtanveer25@student.oulu.fi

Abstract—Emotion recognition systems are increasingly deployed in human-computer and human-robot interaction, yet most multimodal classifiers remain black boxes whose predictions are difficult to audit or trust. This paper presents a tri-modal (text, audio, visual) emotion recognition pipeline built on the Multimodal EmotionLines Dataset (MELD), combining frozen pretrained encoders (RoBERTa for text, Wav2Vec2 for audio, and a Vision Transformer fine-tuned for facial expression recognition) with lightweight trainable fusion classifiers. Beyond classification accuracy, the central contribution of this work is a controlled study of *explanation reliability*: we compare two independent post-hoc attribution methods – exact Shapley values over the three modalities, and Integrated Gradients – against an intrinsic, model-native signal, across two fusion architectures. We find that a simple concatenation-based fusion classifier, combined with per-sample L_2 -normalized embeddings, produces genuinely informative and only partially agreeing explanations (70% top-modality agreement between Shapley and Integrated Gradients, with Integrated Gradients favouring the *visual* modality overall), at a weighted-F1 of 0.566. An attention-gated fusion classifier under otherwise comparable settings reaches a higher weighted-F1 of 0.572–0.578, but its learned gate collapses to assigning 90–98% of its weight to the text modality regardless of input, producing explanation agreement as high as 95–100% – a figure that reflects biased consensus rather than genuine corroboration. We show that this gate collapse additionally distorts Integrated Gradients through the gate’s saturating nonlinearity, inflating text’s apparent importance independent of its true contribution, an effect absent in the gate-free architecture. This leads us to revise an earlier working hypothesis: text dominance is not an intrinsic property of the MELD dataset, but substantially an artifact of attention-gated fusion. Our best configuration approaches the weighted-F1 reported for the context-aware DialogueRNN baseline (0.6025) despite modelling no inter-utterance context, and exceeds it on the surprise class. A locally hosted LLM translates attribution scores into natural-language rationales, illustrating a concrete path from quantitative attributions to human-readable explanations.

Index Terms—multimodal emotion recognition, explainable AI, MELD, Shapley values, Integrated Gradients, attention fusion, focal loss, affective computing.

I. INTRODUCTION

AFFECTIVE computing systems that recognize human emotion from multiple channels – what is said, how it is said, and how it looks – have applications ranging from empathetic dialogue agents to clinical decision support, where

a system’s output must be not only accurate but *explicable* to a non-expert user. This tension between predictive performance and explanation reliability is the central theme of this work.

We build and evaluate a multimodal emotion recognition system on the Multimodal EmotionLines Dataset (MELD) [1] [2], a benchmark of 13,708 utterances from 1,433 multi-party dialogues drawn from the television series *Friends*, each labelled with one of seven emotions and annotated with synchronized text, audio, and video. Rather than treating explainability as an afterthought applied to a single, fixed model, we treat the *choice of embedding normalization, loss function, and fusion mechanism* as variables that jointly determine both classification performance and the trustworthiness of the resulting explanations, and report what happens to each as those choices change.

The contributions of this report are:

- 1) A tri-modal emotion recognition pipeline on MELD built from modern pretrained encoders, extending the original MELD baselines – which used only text and audio – with a visual modality.
- 2) A controlled comparison of two embedding-normalization strategies, showing a reproducible trade-off between weighted-F1 and cross-method explanation agreement.
- 3) A comparison of class-weighted cross-entropy and focal loss, including an ablation over the aggressiveness of the class-weighting term.
- 4) A comparative study of three fusion architectures – simple concatenation, an unconstrained softmax-gated attention fusion, and a hard-floor-constrained variant – that identifies and diagnoses a *gate collapse* failure mode, and shows this failure mode distorts Integrated Gradients attributions through the gate’s own nonlinearity.
- 5) A revised, evidence-based account of why text appears to dominate the other two modalities: we show this is substantially an architectural artifact of gated fusion rather than an intrinsic property of the dataset.
- 6) An LLM-based explanation layer converting quantitative modality-attribution scores into short natural-language rationales for individual predictions.

II. RELATED WORK

MELD [2] [3] extends the text-only EmotionLines corpus with synchronized audio and video, and reports several base-

M. Faizan Tanveer is a second-year master’s student in the Applied Computing track, Computer Science and Engineering, Faculty of Information Technology and Electrical Engineering, University of Oulu, Oulu, Finland.
Manuscript prepared for the Affective Computing course project, 2026.

TABLE I: MELD Dataset Splits

	Train	Dev	Test
Dialogues	1039	114	280
Utterances	9989	1109	2610
Usable utterances	9988	1108	2610
Neutral (%)	47.2	42.4	48.1
Disgust (%)	2.7	2.0	2.6
Fear (%)	2.7	3.6	1.9

lines – a context-free text-CNN, a hierarchical bcLSTM, and DialogueRNN [4], which tracks per-speaker state across a dialogue via gated recurrent units. DialogueRNN remains the strongest published baseline using text and audio, reaching a weighted-F1 of 0.6025 on 7-class emotion recognition; none of the original MELD baselines incorporate the visual modality, which the original authors attribute to the difficulty of speaker localization in multi-party video.

Our interpretability approach is inspired by VISTANet [5], which proposes an adaptive, interpretable fusion of image, speech, and text modalities together with a K-Average Additive exPlanation (KAAP) technique for attributing predictions to individual modalities. We implement an exact Shapley-value variant of this idea, tractable here because the coalition space over three modalities contains only $2^3 = 8$ subsets. As a second, independently-derived attribution method we use Integrated Gradients [7], which attributes a prediction to input features by integrating gradients along a path from a baseline to the actual input. For the imbalance-aware loss function we adopt focal loss [6], originally proposed for dense object detection, which down-weights the loss contribution of confidently-correct examples to concentrate gradient on harder, typically minority-class, examples.

III. DATASET

Table I summarizes the MELD splits used in this work. Two data-quality issues were identified during preprocessing and handled transparently: one training clip (dialogue 125, utterance 3) is corrupted and could not be decoded; one development-split clip (dialogue 110, utterance 7) is absent from the official release, a previously documented omission in the MELD distribution [1]. Both were excluded from their respective splits.

IV. METHODOLOGY

A. Feature Extraction

Each utterance is encoded independently into three fixed-length embeddings, with all encoders kept frozen:

- **Text:** the transcript is tokenized and passed through RoBERTa-base; final hidden states are masked mean-pooled across tokens to a 768-dimensional vector.
- **Audio:** the audio track is extracted from the raw video (16 kHz mono) and passed through Wav2Vec2-base (self-supervised, no ASR head); hidden states are mean-pooled over time to a 768-dimensional vector.
- **Visual:** five frames are sampled uniformly from the clip; a face is localized via MTCNN in each, falling back to the

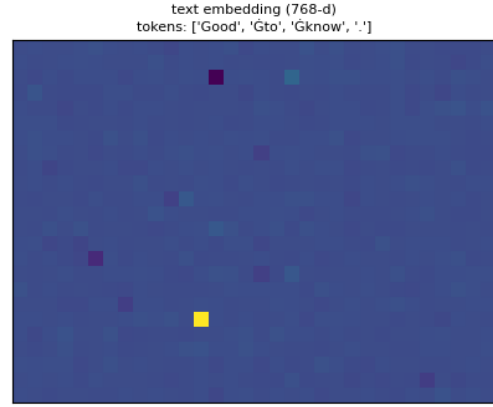


Fig. 1: Text pipeline diagnostic: the 768-dimensional RoBERTa embedding for one utterance, reshaped as a 24×32 heatmap purely for visual inspection, together with the tokenizer’s output.

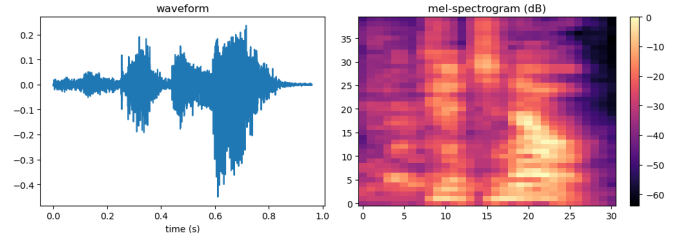


Fig. 2: Audio pipeline diagnostic: raw waveform (left) and mel-spectrogram in dB (right) for one utterance’s extracted audio segment.

full frame when no face is detected (a known challenge in multi-party scenes, per [2]); each crop is passed through a Vision Transformer fine-tuned on FER2013, and CLS-token hidden states are averaged across frames to a 768-dimensional vector.

Each utterance’s embeddings and label are cached to disk once, so all subsequent experimentation operates on a corpus of small tensors rather than the original video.

B. Qualitative Pipeline Diagnostics

Before any model was trained, each stage of the extraction pipeline was verified qualitatively on individual utterances. Fig. 1 visualizes the 768-dimensional text embedding for a short utterance as a reshaped heatmap, alongside the RoBERTa tokenization of the input, confirming a non-degenerate output vector. Fig. 2 shows the corresponding waveform and mel-spectrogram, confirming that the extracted audio segment contains genuine, temporally-structured speech content rather than silence or corrupted samples. Fig. 3 shows a sampled frame with the MTCNN-detected face region, the crop passed to the Vision Transformer, and that model’s own FER2013 class probabilities for one example utterance.

As a complementary, corpus-wide sanity check, Fig. 4 projects all cached embeddings for each modality onto their



Fig. 3: Visual pipeline diagnostic: detected face region (left), the crop fed to the FER2013-finetuned ViT model (centre), and that model’s own predicted emotion-class probabilities (right), for one sampled frame.

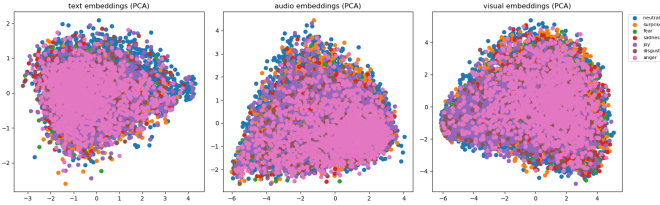


Fig. 4: PCA projection (first two components) of cached text, audio, and visual embeddings across the full dataset, coloured by ground-truth emotion label.

first two principal components, coloured by ground-truth emotion label. No modality shows clean linear separability in two dimensions under any configuration – expected, since none of the three encoders was fine-tuned for emotion classification, and PCA captures only the two directions of greatest overall variance rather than any learned, emotion-discriminative combination of the full 768 dimensions. This figure’s role is to confirm that well-formed, varied embeddings were produced for every utterance, not to demonstrate class separability, which is the fusion classifier’s task.

C. Embedding Normalization Strategies

Table II reports the mean L_2 norm of each modality’s raw embeddings, measured over the training split, and under the two normalization strategies compared in this work.

Text’s raw embeddings carry a norm roughly $2\times$ that of audio and visual. This matters for two reasons: a fusion classifier’s effective sensitivity to each modality is confounded with raw input magnitude, and Integrated Gradients’ attribution, being a gradient-times-input quantity, is directly sensitive to input scale independent of true informativeness. **Per-sample L_2 -normalization** divides every embedding by its own norm, removing all magnitude information – both the across-modality scale gap and each utterance’s own magnitude variation within a modality. **Global per-modality rescaling** instead divides by one fixed constant per modality (its average training-set norm), equalizing average scale across modalities while preserving each utterance’s own relative magnitude variation – for instance, a louder or more energetic speech signal retains a correspondingly larger audio embedding norm. The residual non-zero standard deviations under global rescaling (0.04/0.07/0.17) confirm that genuine per-sample variation is retained, unlike under strict per-sample normalization where it is necessarily zero by construction.

TABLE II: Mean Embedding Norm by Modality and Normalization Strategy

	Text	Audio	Visual
Raw embeddings	12.04 ± 0.52	6.38 ± 0.46	5.29 ± 0.90
Per-sample L_2 -norm	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
Global rescaling	1.00 ± 0.04	1.00 ± 0.07	1.00 ± 0.17

D. Fusion Architectures

Three fusion mechanisms are compared.

Linear (concatenation) fusion concatenates the three embeddings into a single 2304-dimensional vector, passed through a two-layer MLP classifier. As a static, whole-model attribution signal, we additionally compute the L_2 norm of the first classifier layer’s weight sub-matrix corresponding to each modality, normalized to sum to one; unlike the two attribution methods described below, this quantity does not depend on the specific input and is therefore a dataset-level rather than per-example signal.

Soft attention fusion projects each modality into a shared 256-dimensional space and uses a gating network, conditioned on all three projected modalities, to compute an unconstrained softmax weighting over the three before combining them with a weighted sum.

Hard-floor attention fusion is identical to soft attention fusion except that the softmax output is affinely transformed, $w_i \leftarrow w_{\min} + (1 - n w_{\min}) w_i^{\text{raw}}$, guaranteeing every modality weight lies in $[w_{\min}, 1 - (n - 1)w_{\min}]$ (here $w_{\min} = 0.1$, $n = 3$) regardless of the gate’s raw output.

E. Loss Functions

Class-weighted cross-entropy uses the per-class weights reported in the original MELD release (ordered [neutral, surprise, fear, sadness, joy, disgust, anger]: [4.0, 15.0, 15.0, 3.0, 1.0, 6.0, 3.0]). Focal loss [6] additionally down-weights easy examples via $(1 - p_t)^\gamma$, where p_t is the predicted probability of the true class. The standard $\gamma = 2$ setting, combined with the raw class weights as α , underperformed plain weighted cross-entropy in our experiments – a negative result attributed to double-counting the imbalance correction, since the $15\times$ spread in α and focal loss’s own hard-example focusing push in the same direction. Dampening the spread via $\alpha_i \leftarrow \sqrt{\alpha_i}$ (reducing the effective spread from $15\times$ to $\approx 3.9\times$) and lowering γ to 1 substantially closed this gap, and is used in all results reported below.

F. Interpretability Methods

Three signals are computed for each prediction: (i) exact Shapley values over the eight modality coalitions, with a modality “removed” by zeroing its embedding and the characteristic function defined as the model’s softmax probability for the predicted class; (ii) Integrated Gradients [7], computed via Captum with an all-zero baseline, aggregated into per-modality scores by summing absolute attribution within each modality’s slice of the input; and (iii) an intrinsic, architecture-dependent

signal – the gate’s own softmax output for attention-based fusion, or the static weight-norm quantity described above for linear fusion.

G. LLM-based Natural Language Explanations

A locally hosted LLM (Llama 3.2, served via Ollama) is prompted with the utterance transcript, the model’s predicted emotion, and the normalized Shapley (or Integrated Gradients) attribution percentages, and asked to produce a short, plain-language rationale without inventing details not present in the provided context.

V. EXPERIMENTAL SETUP

All models are trained with AdamW (learning rate 1×10^{-3} , weight decay 1×10^{-4}), batch size 8, for up to 20 epochs with early stopping (patience 4) on development-set weighted-F1, using a fixed random seed across compared runs. Two final configurations are reported in detail:

Case 1 (linear fusion): per-sample L_2 -normalized embeddings, no modality dropout, $\sqrt{\alpha}$ class weighting, $\gamma = 1$.

Case 2 (soft attention fusion, no entropy penalty): global per-modality rescaling, modality dropout ($p = 0.15$), $\sqrt{\alpha}$ class weighting, $\gamma = 1$.

VI. RESULTS AND DISCUSSION

A. Classification Performance

Table III reports weighted-F1 for both cases under both loss functions, against the DialogueRNN baseline.

TABLE III: Weighted-F1 on the MELD Development Split

Configuration	Cross-Entropy	Focal ($\gamma=1$)
Initial baseline (no norm., raw α)	0.517	–
Case 1: Linear, L_2 -norm	0.566	0.554
Case 2: Soft attention, rescaled	0.583	0.578
DialogueRNN (text+audio) [4]	0.6025	

Table IV gives the per-class F1 breakdown for both cases against DialogueRNN.

TABLE IV: Per-Class F1 Comparison

Class	Case 1 (CE)	Case 2 (Focal)	DialogueRNN
Neutral	0.734	0.730	0.774
Surprise	0.579	0.609	0.525
Fear	0.095	0.225	0.117
Sadness	0.273	0.325	0.346
Joy	0.538	0.555	0.544
Disgust	0.000	0.057	0.079
Anger	0.483	0.457	0.437

Our best configuration (Case 2, cross-entropy loss) comes within 2 points of DialogueRNN’s weighted-F1 despite modelling no inter-utterance context, and exceeds it on the surprise class in every configuration tested, which we attribute to the inclusion of the visual modality, absent from all original MELD baselines. Fear and disgust remain difficult across

every model compared, including DialogueRNN, suggesting these are genuinely hard classes for this dataset and feature representation.

Figs. 5–12 show training curves and development-set confusion matrices for all four Case 1/Case 2 $\times$ CE/Focal combinations.

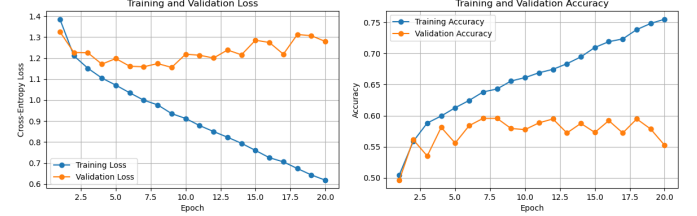


Fig. 5: Case 1 (linear fusion), cross-entropy loss: train/validation loss and accuracy.



Fig. 6: Case 1 (linear fusion), focal loss ($\gamma=1$): train/validation loss and accuracy.

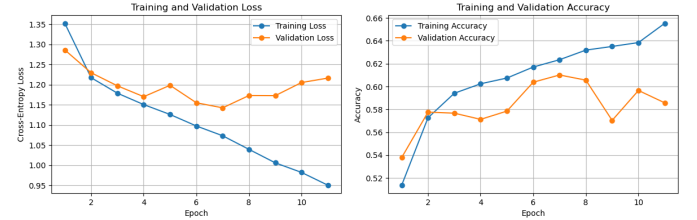


Fig. 7: Case 2 (soft attention fusion), cross-entropy loss: training/validation loss and accuracy.

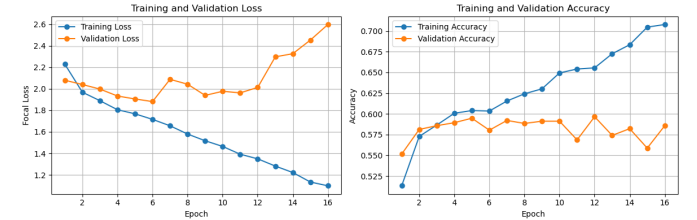


Fig. 8: Case 2 (soft attention fusion), focal loss ($\gamma=1$): training/validation loss and accuracy.

Across all four combinations, validation loss begins diverging from training loss within the first 3–5 epochs while validation accuracy plateaus, consistent with the limited capacity of a frozen-encoder, lightweight fusion classifier to keep extracting new signal from $\sim 10k$ training utterances; early stopping on weighted-F1 (Section V) selects a checkpoint well before this divergence becomes severe.

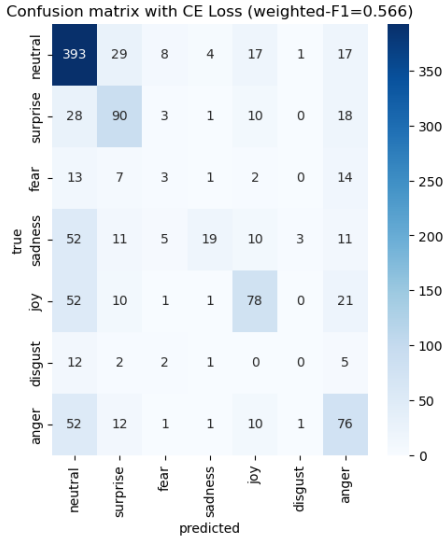


Fig. 9: Case 1 (linear fusion), cross-entropy loss: development-set confusion matrix (weighted-F1 = 0.566).

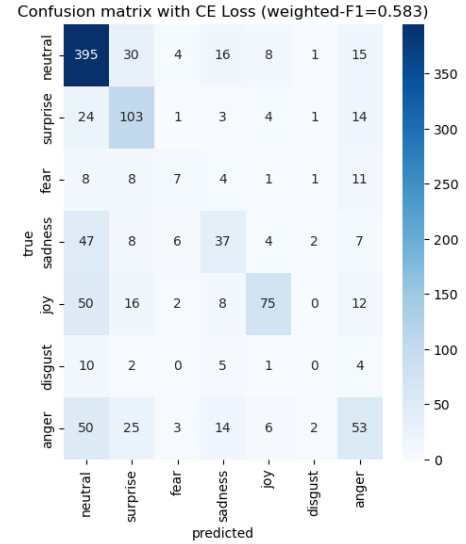


Fig. 11: Case 2 (soft attention fusion), cross-entropy loss: development-set confusion matrix (weighted-F1 = 0.583).

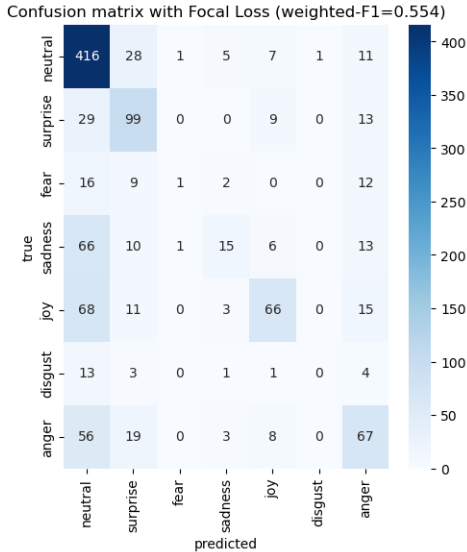


Fig. 10: Case 1 (linear fusion), focal loss: development-set confusion matrix (weighted-F1 = 0.554).

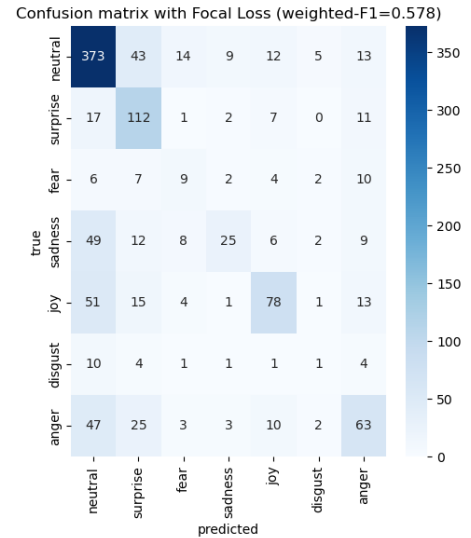


Fig. 12: Case 2 (soft attention fusion), focal loss: development-set confusion matrix (weighted-F1 = 0.578).

Every confusion matrix shows the same qualitative pattern: neutral and surprise are classified with high recall, while sadness, fear, and especially disgust leak heavily into neutral, mirroring the per-class trends in Table IV and the equivalent difficulty reported for DialogueRNN.

B. Accuracy versus Explainability: Linear versus Attention Fusion

Case 1 and Case 2 differ in both fusion architecture and normalization strategy, isolating a combined effect we consider jointly representative of a practical deployment choice. Case 2 achieves 0.6–2.9 points higher weighted-F1 than Case 1. However, Case 2’s attention gate assigns 90–98% of its weight to the text modality for both loss functions, regardless of the specific utterance, while Case 1’s linear fusion shows no such

collapse: its static weight-norm signal and, more importantly, its two per-example attribution methods both show meaningful variation across utterances (Section VI-D). Top-modality agreement between Shapley and Integrated Gradients is 70% for Case 1 under both losses, compared to 95% (CE) and 75% (focal) for Case 2. We interpret Case 2’s higher agreement not as evidence of more trustworthy explanations, but largely as a mechanical consequence of there being only one modality left for either method to point to.

C. Shapley versus Integrated Gradients: Two Lenses, Sometimes Very Different Views

Under Case 1 (CE loss), Shapley attributes an average of 44% of the model’s confidence to text (with a *negative* average contribution from audio, −31%, indicating audio evidence on

average argued against the model’s own prediction), while Integrated Gradients attributes the plurality of its gradient-based signal to the *visual* modality (39%, versus 36% text and 25% audio). This is a substantive, non-trivial disagreement between two independently-derived methods applied to the same model and the same predictions – precisely the kind of result that justifies running more than one attribution method. Fig. 13 and Fig. 14 show the resulting per-utterance top-modality agreement under Case 1 for both loss functions (70% in both cases, but with different confusion patterns between methods).

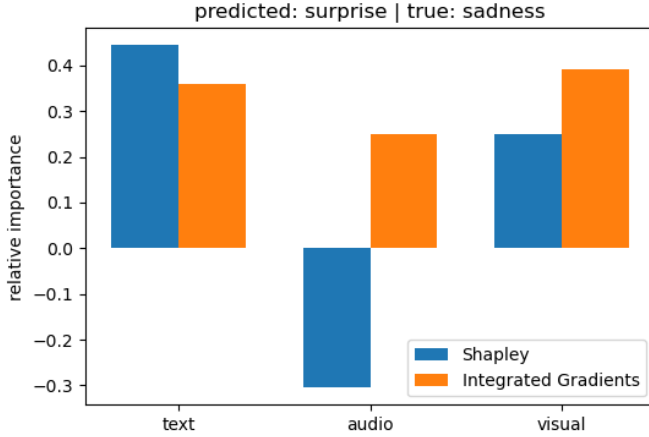


Fig. 13: Case 1, cross-entropy loss: Shapley vs. Integrated Gradients attribution for a representative utterance, and top-modality agreement across 10 sampled development-set utterances (70%).

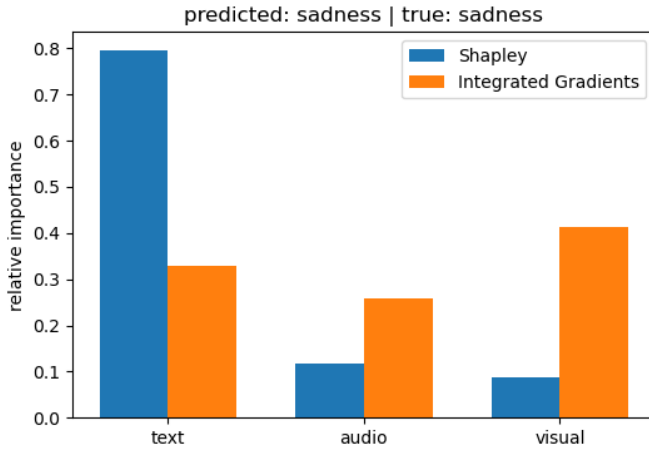


Fig. 14: Case 1, focal loss: Shapley vs. Integrated Gradients attribution for a representative utterance, and top-modality agreement across 10 sampled development-set utterances (70%).

Under Case 2, by contrast, both methods converge almost entirely on text (93–98%), which Section VI-E argues is attributable to the fusion architecture rather than to the two methods independently corroborating a genuine signal.

D. Gate Collapse in Attention Fusion

Across every attention-fusion configuration tested in this project – with and without modality dropout, with and without a hard floor on gate weights, with and without an entropy penalty discouraging low-entropy gate outputs – the learned gate converges to assigning the overwhelming majority of its weight to the text modality. A hard floor constraining weights to $[0.1, 0.8]$ does not prevent this: the underlying softmax still saturates completely, and the floor transform merely relabels a fully collapsed $[1, 0, 0]$ output as $[0.8, 0.1, 0.1]$ without altering the gate’s actual decision process. We attribute the collapse to a “rich-get-richer” training dynamic: because text is the most informative single modality for the majority of utterances, the gate discovers this early in training, and because silencing a modality costs the optimizer only a handful of gate parameters – as opposed to the hundreds of individual weights a concatenation-based classifier would need to drive toward zero to achieve the same effect – nothing in the architecture resists a runaway collapse once it begins. Figs. 15, 16 show the corresponding attribution comparison and agreement statistics.

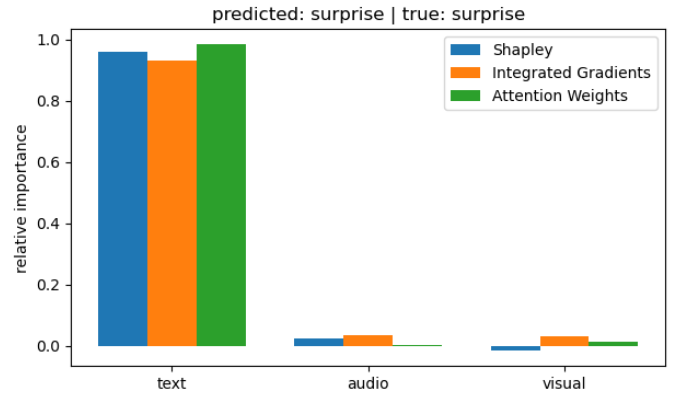


Fig. 15: Case 2, Cross-entropy loss: Shapley vs. Integrated Gradients vs. the gate’s own attention weights; three-way agreement across 20 sampled utterances is 95%.

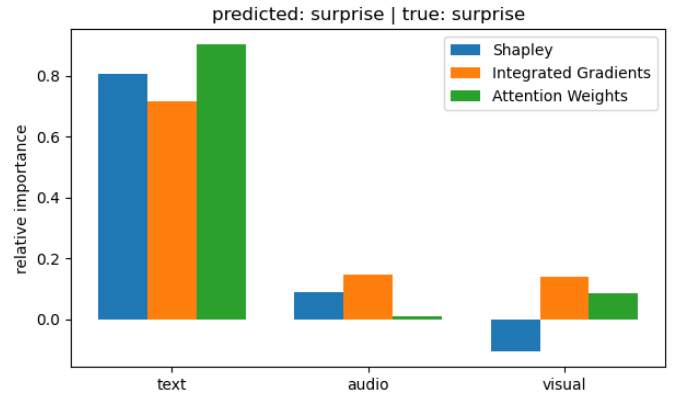


Fig. 16: Case 2, focal loss: Shapley vs. Integrated Gradients vs. the gate’s own attention weights; three-way agreement across 20 sampled utterances is 70%, lower than under cross-entropy because the focal-trained gate is marginally less saturated.

E. Architecture-Dependence of Integrated Gradients

Comparing Integrated Gradients’ attributions between Case 1 and Case 2 under cross-entropy loss reveals a striking asymmetry: under linear fusion, IG attributes only 36% to text and the plurality (39%) to visual; under soft attention fusion, IG attributes 93% to text. Both models explain predictions on the same dataset with comparable loss functions. We interpret this as evidence that the gate’s saturating softmax nonlinearity distorts Integrated Gradients’ path-integral computation: as the integration path crosses the region where the gate’s output saturates, a disproportionate share of the attributed gradient concentrates there, inflating the apparent importance of whichever modality the gate favours, independent of that modality’s true marginal contribution as measured by Shapley’s ablation-based approach. To our knowledge this is a previously undocumented interaction between gated multimodal fusion and gradient-based attribution, and we recommend it be treated as a live methodological concern in future work combining learned fusion gates with gradient-based explanation methods.

F. Is Text Dominance a Finding or an Artifact? A Revised Account

An earlier iteration of this investigation, based primarily on attention-gated architectures, concluded that text dominance was likely an intrinsic property of the MELD dataset and feature-extraction pipeline. The Case 1 results revise this conclusion. Under a gate-free, linear fusion architecture, Integrated Gradients does not favour text at all – it favours visual – and Shapley’s attributions, while still text-leaning on average, show meaningful per-utterance variation rather than uniform convergence. We therefore conclude that **the appearance of near-universal text dominance is substantially an artifact of attention-gated fusion, not a property of the underlying data**. This revision illustrates a broader methodological point: claims about which modality “matters most” in a multimodal system should be checked against more than one fusion architecture before being reported as a property of the data itself.

G. Qualitative Case Studies

Table V contrasts LLM-generated explanations for the same utterance, “What?”, across Case 1 and Case 2.

TABLE V: LLM Explanations for the Same Utterance Under Different Configurations

Utterance: “What?” (Monica) — Predicted/True: surprise/surprise
Case 1, Shapley-based: “...her tone of voice, which is conveyed through the audio modality, is quite abrupt and inquiring... the audio contribution was the primary driver of the surprise prediction, outweighing the more neutral language in her words.”
Case 1, IG-based: “...the text modality, specifically the words she chose, played a significant role in this decision. The phrase ‘What?’ is often used as an expression of surprise or inquiry...”
Case 2, Shapley-based: “...the text contribution is 81%, suggesting that the words ‘What?’ are crucial in conveying a sense of questioning or inquiry...”

Under Case 1, Shapley-based and Integrated-Gradients-based explanations genuinely differ in their account of the same prediction – one citing vocal tone, the other citing the lexical content itself – reflecting the real divergence in their underlying numbers. Under Case 2, the explanation is dominated by text regardless of which attribution method supplies the input, consistent with the gate collapse documented above.

VII. LIMITATIONS

- Interpretability metrics were computed on a sample of 10–20 development-set utterances per configuration for computational tractability; broader claims would benefit from evaluation over the full development set.
- Shapley’s zero-ablation assumption is only partially justified even with modality dropout: the model’s behaviour under ablation combinations never seen during training remains an approximation rather than a fully calibrated counterfactual.
- No inter-utterance conversational context is modelled; every utterance is classified independently of its dialogue history, unlike DialogueRNN.
- The disgust class could not be learned above chance by most configurations tested, consistent with its difficulty in the original MELD baselines, and with only 22 development-set examples we cannot rule out that this is partly a data-volume limitation.

VIII. FUTURE WORK

The most promising direction for closing the remaining gap to context-aware baselines is to model each dialogue as a sequence, following DialogueRNN’s approach of tracking per-speaker state across turns. The gate-collapse phenomenon documented here warrants investigation with a broader family of regularization strategies – for example, load-balancing losses from the sparse mixture-of-experts literature – before attention-based fusion can be recommended over simple concatenation when one modality is substantially more informative than the others. The architecture-dependence of Integrated Gradients identified in Section VI-E merits further study in its own right, potentially extending to other gated or attention-based architectures beyond multimodal fusion. Finally, extending the Shapley/Integrated-Gradients agreement analysis to the full development and test splits, and validating the LLM explanation layer against human judgments of explanation quality, would strengthen the interpretability claims made in this work.

IX. CONCLUSION

This work [8] presented a tri-modal emotion recognition pipeline on MELD and used it as a testbed for studying how representation, loss, and fusion-architecture choices affect both classification accuracy and the reliability of post-hoc explanations. We found a reproducible trade-off between weighted-F1 and cross-method explanation agreement governed by embedding normalization and fusion architecture, documented a gate-collapse failure mode in attention-weighted fusion together with its previously unreported distorting effect

on Integrated Gradients, and used this evidence to revise an earlier conclusion: text dominance in our results is substantially an artifact of gated fusion rather than an intrinsic property of the MELD dataset. A simple concatenation-based fusion classifier offers comparable accuracy with meaningfully more informative and architecture-robust explanations, and is recommended as the primary model where explanation reliability is a project requirement rather than a secondary concern.

ACKNOWLEDGMENT

Large language model assistance (Anthropic Claude) was used throughout this project for code review, debugging, explanation of statistical and deep learning concepts, and drafting assistance for this report. The author made all experimental design decisions, chose the dataset and methodology, implemented the code, and interpreted the results.

REFERENCES

- [1] Declare-Lab, “MELD dataset,” Hugging Face Datasets. [Online]. Available: <https://huggingface.co/datasets/declare-lab/MELD>
- [2] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, “MELD: A multimodal multi-party dataset for emotion recognition in conversations,” in *Proc. ACL*, 2019.
- [3] Declare-Lab, “MELD: Multimodal EmotionLines Dataset – official repository,” GitHub. [Online]. Available: <https://github.com/declare-lab/MELD>
- [4] N. Majumder, S. Poria, D. Hazarika, R. Mihalcea, A. Gelbukh, and E. Cambria, “DialogueRNN: An attentive RNN for emotion detection in conversations,” in *Proc. AAAI*, 2019.
- [5] P. Kumar, S. Malik, and B. Raman, “Hybrid fusion based interpretable multimodal emotion recognition with limited labelled data,” *arXiv:2208.11450*, 2022.
- [6] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. ICCV*, 2017.
- [7] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic attribution for deep networks,” in *Proc. ICML*, 2017.
- [8] “Interpretable Multimodal Emotion Recognition on MELD with LLM-Generated Explanations,” GitHub. [Online]. Available: <https://github.com/FaizanTanwir/AC2026-Interpretable-Multimodal-Emotion-Recognition-on-MELD>